



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	APPLICATIONS OF ADVANCED MACHINE LEARNING & ARTIFICIAL INTELLIGENCE IN PHYSICS				
Type of Course	Vocational Minor (GROUP II: DATA ANALYSIS IN PHYSICS)				
Semester	VIII				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. PHY1VN102- Python Basics 2. PHY2VN102- Data Analysis in Physics Using Python 3. PHY3VN202- Data Analysis in Physics Using Machine Learning				
Course Summary	This course explores the fundamentals of Artificial Intelligence: Basic idea about AI. It also explains the advanced concepts of Machine Learning Techniques. Deep Learning and CNNs are introduced.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Acquire expertise in DBSCAN for spatial clustering and neural networks for comprehensive data analysis and pattern recognition proficiency.	Ap	P	Practical Assignment / Observation of Practical Skills

CO2	Grasp the significance of SVM, apply it using Python, adjust parameters, evaluate pros/cons, and employ it across varied applications.	U	C	Instructor-created exams / Quiz
CO3	Understand the Deep Learning concepts, utilise the TensorFlow/Keras framework, grasp neural network variants, and understand various neural network architectures.	U	C	Seminar Presentation / Group Tutorial Work
CO4	Develop machine learning models for practical applications, enhancing skills in classification, feature selection, and model evaluation techniques.	Ap	P	Instructor-created exams / Home Assignments
CO5	Grasp the concepts and importance of Artificial Intelligence, historical context and how the brain processes information.	U	C	One Minute Reflection Writing assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I		Neural Networks and Clustering Techniques in ML	13	18
	1	Density Based Spatial Clustering of Applications with Noise (DBSCAN) - Understand how DBSCAN works	2	
	2	Algorithmic steps for DBSCAN clustering - parameter estimation	1	
	3	Python implementation of DBSCAN using Scikit-learn - example using random sample generation	3	
	4	Neural Network - Working of a neural network - model - Pros and Cons	3	

	5	Applications of neural networks - Activation Function - Steps involved in neural network methodology - Example using scikit-learn (not for examination)	4	
	Sections 11.5 - 11.5.5 and 11.7 - 11.7.6 of Chapter 11 of Book 1			
II	Support Vector Machine		11	16
	6	Support Vector Machine (SVM) - Need of SVM	2	
	7	Important terms in SVM - Hyperplane - Margin - Tuning Parameters	2	
	8	Working of SVM - Advantages and Disadvantages of SVM	2	
	9	Applications of SVM	2	
	10	Tuning hyperparameters - Python implementation of SVM - Example data using breast cancer (Not for examination)	3	
	Section 11.8 - 11.8.6 of Chapter 11 of Book 1			
III	Advanced Machine Learning Techniques		13	20
	11	Deep Learning - Working of DL Model - Comparison between ML and DL	2	
	12	Applications of Deep Learning - Libraries for implementing DL - TensorFlow and Keras	3	
	13	Types of Neural Networks - ANN - MLP - CNN - RNN	3	
	14	Architecture of Keras - Model - Layer	2	
	15	Loss - Optimizer - Metrics	1	
	16	Training the model - With ionosphere data to identify any structure is present in a radar data using Keras (Not for examination)	2	
	Section 12.1 - 12.4 of Chapter 12 of Book 1			
IV	Foundations of Artificial Intelligence		11	16
	17	What is Artificial Intelligence - Turing Test - Cognitive modeling approach	2	
	18	Foundations of AI - Philosophy	2	
	19	How do brain process information - How can we build an efficient computer	1	
	20	History of AI - The birth - Early Enthusiasm - Availability of large data sets	2	
	21	Knowledge-based systems - AI adopts the scientific method	2	

	22	Intelligent agents -The State of art	2	
	Section 1.1 - 1.4 of Chapter 1 of Book 2			
	OPEN ENDED MODULE		12	
V	Implement one of the following tasks or any other relevant project: 1. Photometric Redshift Estimation using the data: Data: Read from Scikit-learn 2. Develop a neural network for the detection of exoplanet: Data: Repository given in the reference section 3. Develop a SVM model for the detection of exoplanet: Data:Repository given in the reference section			
	1. https://ogrisel.github.io/scikit-learn.org/sklearn-tutorial/tutorial/astromy/regression.html 2. https://github.com/gabrielgarza/exoplanet-deep-learning/tree/master			

Books of Study:

- Data Science and Machine Learning using Python by Reema Thereja
- Artificial Intelligence – A Modern Approach Third Edition by Stuart Russel and Peter Norvig.

Reference:

1. Machine Learning in Data Science using Python by R Nageswara Rao

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	1	1	2	1	1	1	1	2	2	2	1	1
CO 2	1	2	2	1	1	1	1	1	2	2	2	1	1
CO 3	1	1	3	1	2	1	1	1	2	3	3	1	1
CO 4	1	2	3	3	1	1	1	1	3	3	3	1	1
CO 5	1	1	1	1	3	1	2	1	1	2	1	1	1
CO 6	2	1	1	2	1	1	1	1	2	2	2	1	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	

GENERAL FOUNDATION COURSES